DATA ANALYTICS WITH POWER BI

GLOBAL MUSIC TRENDS AND POPULARITY ANALYTICS

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Declaration

I, Mahalakshmi S(12323862), hereby declare that the project work entitled **"Global Music Trends and Popularity Analytics"** is an authentic record of my own work carried out under the supervision of **Dr. Ashu Mehta**. This report has not been submitted to any other University or Institute for the award of any degree.

Signature: Mahalakshmi S

Date: 21-12-2025

Certificate

This is to certify that the project report entitled **"Global Music Trends and Music Analytics"** submitted by **Mahalakshmi S** in partial fulfillment of the requirements for the Minor Project in Data Science is a record of work carried out by them under my guidance and supervision.

Dr. Ashu Mehta

Department of Computer Science

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ACKNOWLEDGEMENT

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I would like to acknowledge **OpenDataBay** for providing access to a comprehensive and reliable dataset, which served as the foundation of this analytical study.

Finally, I express my sincere appreciation to my family and friends for their unwavering support and motivation during the course of this project.

Mahalakshmi S

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1. Introduction

In recent years, the rapid growth of digital music streaming platforms has led to the generation of massive volumes of music-related data. Platforms such as Spotify collect detailed information not only about songs and artists but also about a wide range of audio features, listener preferences, and popularity metrics. Analyzing such data using data science and business intelligence tools helps uncover meaningful patterns that explain what makes a song popular, how genres differ in their musical characteristics, and how artists perform over time.

This project focuses on performing an in-depth exploratory and analytical study of a large-scale music dataset using **Microsoft Power BI**. The dataset contains information on thousands of tracks, artists, genres, popularity scores, and multiple audio attributes such as danceability, energy, loudness, tempo, valence, acousticness, and instrumentalness. Due to the size and complexity of the dataset, careful consideration was given to selecting visualization techniques that are both **computationally efficient** and **analytically insightful**.

The primary objective of this analysis is to transform raw music data into meaningful insights through interactive dashboards. The study aims to identify the key factors influencing song popularity, understand genre-wise audio characteristics, evaluate artist performance, explore natural groupings of songs based on audio features, and examine relationships between different musical attributes. While several advanced visualization techniques exist, not all are suitable for large datasets in Power BI. Therefore, this project emphasizes the use of **aggregated, AI-driven, and scalable visuals** to ensure performance stability and clarity.

To achieve these goals, five well-defined analytical objectives were framed, each supported by a carefully chosen Power BI visualization. The **Key Influencers visual** was used to analyze popularity drivers, replacing traditional correlation-based charts that were not compatible with the dataset size. For the remaining objectives, performance-friendly native visuals such as **matrix, clustered bar charts, and tree map** were employed instead of complex or custom visuals that could lead to rendering issues.

Overall, this project demonstrates how large music datasets can be effectively analyzed using Power BI by combining data preprocessing, intelligent modeling, and appropriate visualization strategies. The insights derived from this analysis can be valuable for music analysts, streaming platforms, artists, and researchers seeking to understand trends in digital music consumption.

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2. Scope of the Analysis

The scope of this project is centered on extracting meaningful insights from a large-scale music dataset using data science and business intelligence techniques. The analysis is designed to remain **descriptive, diagnostic, and exploratory**, ensuring that conclusions are interpretable, scalable, and suitable for academic evaluation.

2.1 Analytical Scope

This project focuses on analyzing **audio feature metadata** and **popularity indicators** of songs rather than listener-specific behavioral data. The scope includes:

- Studying how **audio characteristics** such as danceability, energy, tempo, loudness, valence, acousticness, and instrumentalness influence song popularity.
- Performing **genre-wise audio profiling** to understand how musical attributes vary across different genres.
- Evaluating **artist performance** based on aggregated popularity metrics.
- Identifying **natural groupings of songs** using audio features to understand mood and style clusters.
- Exploring **relationships between musical attributes** to observe trends and patterns across large datasets.

The analysis is performed using **aggregated measures** rather than individual-level predictions to maintain performance efficiency in Power BI and ensure statistically meaningful insights.

2.2 Technical Scope

From a technical perspective, the scope includes:

- Data preprocessing and transformation using **Power Query Editor**.
- Creation of calculated columns and measures using **DAX (Data Analysis Expressions)**.
- Development of interactive dashboards using **Power BI native and AI visuals**.
- Optimization of visuals to handle a **large dataset without performance degradation**.

Advanced machine learning model deployment and real-time streaming analytics are **outside the scope** of this project and are proposed as future enhancements.

2.3 Tool and Platform Scope

The entire analysis is limited to the following tools and technologies:

- **Microsoft Power BI** for data modeling, analysis, and visualization
- **CSV-based structured dataset** sourced from OpenDataBay
- Built-in Power BI visuals and AI visuals (Key Influencers)

No external scripting languages such as Python or R are used in this phase of the project.

2.4 Functional Scope

The project supports the following functional outcomes:

- Identification of **key drivers of music popularity**
- Comparative analysis of genres and artists
- Visual exploration of high-dimensional audio data
- Interactive filtering and drill-down analysis using slicers

The results are intended to support **data-driven decision-making** in the music analytics domain.

2.5 Limitations of Scope

While the dataset is large and diverse, the scope does not include:

- Listener demographics or user engagement metrics
- Real-time streaming or recommendation systems
- Predictive or prescriptive analytics models

These areas are intentionally excluded to maintain clarity, feasibility, and academic alignment.

3. Existing System

3.1 Overview of the Existing System

The existing system for evaluating music success in the industry is largely **intuition-driven and experience-based**. Traditionally, decisions related to song promotion, artist signing, and genre trends have relied on:

- Manual listening and expert judgment by A&R (Artists and Repertoire) teams
- Historical chart performance and basic streaming counts
- Social media trends and public sentiment
- Limited descriptive statistics provided by streaming platforms

Most analyses focus on **surface-level metrics** such as total streams, likes, or chart rankings, without examining the **underlying audio characteristics** that contribute to a song's popularity.

In academic and small-scale analytical settings, music data is often analyzed using:

- Spreadsheet tools like Microsoft Excel
- Static charts and summary tables
- Small datasets with limited genre or artist diversity

These approaches are sufficient for **basic reporting**, but they fail to scale when dealing with **large, high-dimensional datasets** such as modern music metadata.

3.2 Drawbacks / Limitations of the Existing System

Despite being widely used, the existing system has several critical limitations:

1. Lack of Data-Driven Insights

Traditional methods rely heavily on subjective judgment rather than empirical analysis. This makes it difficult to:

- Identify measurable drivers of popularity
- Compare songs objectively across genres
- Predict trends based on audio features

2. Inability to Handle Large Datasets

Conventional tools struggle with:

- High-volume datasets (100,000+ records)
- Multiple continuous audio features
- Real-time filtering and comparison

As a result, insights are often derived from **small samples**, leading to biased conclusions.

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3. No Multidimensional Analysis

Existing systems typically analyze one metric at a time (e.g., popularity or streams). They do not support:

- Simultaneous analysis of multiple audio features
- Pattern detection across dimensions such as mood, rhythm, and energy
- Visual clustering or comparative profiling

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4. Static and Non-Interactive Reporting

Most reports generated using traditional methods are:

- Static in nature
- Non-interactive
- Unable to adapt to user-selected filters such as genre or artist

This limits exploratory analysis and decision-making flexibility.

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5. Absence of Advanced Analytics

The existing system lacks:

- AI-driven insights (e.g., key influencing factors)
- Automated trend detection
- Diagnostic analytics explaining *why* a song performs well

This results in analysis that explains **what happened**, but not **why it happened**.

3.3 Need for an Improved System

Due to these limitations, there is a strong need for a **modern, scalable, and analytical system** that can:

- Process large-scale music datasets efficiently
- Identify hidden patterns using audio features
- Provide interactive, visual-driven insights
- Support evidence-based decision-making

This project addresses these gaps by leveraging **Power BI, DAX, and advanced visualization techniques** to build a robust music analytics framework.

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4. Source of Dataset

4.1 Dataset Overview

The dataset used for the **"Global Music Trends and Popularity Analytics"** project was sourced from **OpenDataBay**, a public data repository known for hosting large-scale, high-quality datasets suitable for academic and analytical research.

The dataset is titled **"Spotify Track Audio Features and Genre Dataset"** and contains detailed metadata and audio characteristics of music tracks collected from the Spotify platform.

This dataset was chosen because it goes beyond basic information such as artist name and track title, and instead provides a **comprehensive digital audio profile** for each song. Such depth is essential for performing meaningful data science analysis on music popularity.

4.2 Dataset Size and Structure

- **Number of Records:** 114,000 tracks
- **Number of Columns:** 15
- **File Format:** CSV (Comma-Separated Values)
- **Data Type:** Structured, tabular dataset
- **Granularity:** Track-level data

Each row in the dataset represents **one unique music track**, while each column represents a specific attribute related to that track.

4.3 Key Attributes in the Dataset

The dataset includes a diverse range of variables, broadly categorized as follows:

1. Track Metadata

- Track ID (unique identifier)
- Track Name
- Artist Name
- Album Name
- Track Genre

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2. Audio Features

These features are extracted directly from Spotify's audio analysis algorithms:

- **Danceability:** Measures how suitable a track is for dancing
- **Energy:** Represents intensity and activity level
- **Loudness:** Overall loudness in decibels (dB)
- **Speechiness:** Presence of spoken words
- **Acousticness:** Likelihood of the track being acoustic
- **Instrumentalness:** Probability of the track containing no vocals
- **Liveness:** Detects presence of a live audience
- **Valence:** Musical positiveness conveyed by a track
- **Tempo:** Beats per minute (BPM)
- **Time Signature:** Estimated beats per measure
- **Key and Mode:** Musical scale and modality

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3. Popularity Metric

- **Popularity:** A numeric score ranging from 0 to 100, representing the commercial success of a track based on Spotify's internal metrics.

4.4 Reliability and Relevance of the Dataset

The dataset is highly reliable for analysis due to:

- **Large sample size**, reducing statistical bias
- **Standardized feature extraction**, ensuring consistency
- **Global coverage**, including multiple genres and artists
- **Industry relevance**, as Spotify is a leading music streaming platform

The presence of both **quantitative audio features** and **categorical genre information** makes this dataset ideal for:

- Exploratory Data Analysis (EDA)
- Pattern recognition
- Popularity trend analysis
- Dashboard-driven storytelling

4.5 Justification for Dataset Selection

This dataset was selected because it:

- Enables multidimensional analysis of music
- Supports advanced analytics in Power BI
- Aligns with the project's objective of understanding popularity drivers
- Is suitable for large-scale visualization without excessive aggregation loss

I completely understand! You used the **user-friendly interface** in Power BI (the "Conditional Column" and "Transform" buttons) rather than writing code by hand. This is actually the standard professional way to do it.

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5. Data Preprocessing and Cleaning

5.1 Preprocessing Objectives

The transformation of 114,000 raw records from **OpenDataBay** into a high-performance dashboard required a structured cleaning process within the **Power Query Editor**. The goal was to take raw API metadata and refine it into "Analysis-Ready" information. This was done using built-in transformation tools to ensure data uniformity and to create new categories that make the final charts easier to read.

5.2 Technical Cleaning and Transformation Steps

5.2.1 Text Standardization (Track Genre)

A major issue with large datasets is inconsistent text (e.g., "pop" vs "Pop"). To fix this, I applied a **Text Transformation** to the Track_Genre column.

- **Action:** Used the **"Capitalize Each Word"** (Text.Proper) function.
- **Result:** This ensured that every genre follows a standard naming convention, allowing the "Genre Slicer" on the dashboard to be clean and professional.

5.2.2 Temporal Normalization (Duration Conversion)

The raw data provided track lengths in milliseconds (e.g., 240000 ms), which is difficult for a user to interpret.

- **The Process:** I used the **"Standard"** math transformation to divide the column by **60,000**.
- **Refinement:** I then used the **"Rounding"** tool to set the values to **2 decimal places**.
- **Result:** The dashboard now displays song lengths in "Minutes," which is much more intuitive for music industry analysis.

5.2.3 Feature Engineering: The "Energy Bin" Logic

To allow for better clustering in our AI visuals, I used the **"Conditional Column"** tool to create an **Energy_Bin**. Instead of looking at a decimal like 0.82, the dashboard can now group tracks by their "Intensity Level."

- **The Logic:**
    - **Very High:** Energy > 0.8
    - **High:** Energy > 0.6
    - **Medium:** Energy > 0.4
    - **Low/Very Low:** Energy < 0.4

5.2.4 Rhythmic Categorization (Tempo)

Similarly, I created a **Tempo_Category** to simplify the analysis of track speed. This helps identify if "Fast" songs are more popular than "Slow" songs.

- **The Logic:**
    - **FAST:** Above 130 BPM
    - **MEDIUM:** Between 90 and 130 BPM
    - **SLOW:** Below 90 BPM

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6. Data Modelling

6.1 Model Architecture

The project utilizes a **Flat-Table Schema** optimized. By processing all 114,000 rows in a single table, the model is built for speed. All the "Bins" and "Categories" created during the cleaning phase were loaded into the model as primary dimensions for slicing and dicing the data.

6.2 Measure Engineering (DAX)

Once the data was cleaned, I used **DAX (Data Analysis Expressions)** to create the core metrics that drive the visualizations. These measures are dynamic and recalculate based on the user's selection.

6.2.1 Primary Aggregate Measures

- **Total Track Count:** A COUNTROWS function was used to keep track of the 114,000 records. This is vital for showing the "Density" of specific genres.
- **Average Popularity:** An AVERAGE function applied to the popularity score to provide a benchmark for success.

- **Average Audio Features:** Separate measures were created to find the Mean (Average) for **Acousticness, Energy, and Valence** across the entire dataset.

6.2.2 Categorical Distribution

By using the new **Energy_Bin** and **Tempo_Category** in our model, we can now perform "Cross-Highlighting." For example, a user can click on "FAST" music and instantly see the **Average Popularity** and **Track Count** for that specific speed category.

6.3 Data Type Optimization

To ensure the report remains fast, I performed a final "Data Type" check in the Model View:

- **Decimal Numbers:** For Energy, Valence, and Duration.
- **Whole Numbers:** For Popularity and Tempo.
- **Text:** For the new Categories (Bins) and Genres.

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7. Analysis on Dataset

This section details the analytical processes executed to derive meaningful insights from the 114,000 tracks. Each analysis follows a structured methodology to explain the "what," "how," and "why" behind the musical trends observed.

7.1 Analysis 1: Popularity Drivers Analysis (AI-Powered)

7.1.1 Introduction

The core objective of this analysis is to move beyond simple correlations and identify the specific audio characteristics that serve as primary drivers for a track's commercial success. This diagnostic approach answers the fundamental question: *"What technical attributes statistically increase a song's likelihood of becoming a hit?"*

7.1.2 General Description

Manual observation of 114,000 records is insufficient to detect complex patterns. Therefore, this analysis utilizes the **Key Influencers AI visual**. By applying machine learning models internally, the system identifies how changes in independent variables (audio features) influence the average of the dependent variable (Popularity).

7.1.3 Specific Requirements, Functions, and Formulas

- **Analyze Field:** Popularity
- **Explain By Fields:** Acousticness, Danceability, Instrumentalness, and Liveness.
- **Feature Engineering:** The visual automatically bins continuous data to find the exact thresholds (e.g., Acousticness ≤ 0.504) where popularity shifts occur.

7.1.4 Analysis Results

- **Acousticness Threshold:** Popularity is significantly more likely to increase when **Acousticness is 0.504 or less**. This results in an average popularity increase of approximately **20.61 units**.
- **Vocal Content:** High vocal presence (low instrumentalness) is a major driver, with popularity increasing by **15.02 units** when instrumentalness is extremely low (below 4.13E-5).
- **Rhythmic Preference:** A danceability score of **0.57 or less** (in this specific filtered context) and low liveness (**0.12 or less**) also showed marked influence on popularity spikes.

7.1.5 Visualization

- **Visual Used:** Key Influencers (AI Visual).
- **Why this visual was chosen:** It provides automated cause-and-effect insights that are statistically validated, offering a high-level summary of "Key Drivers" for decision-makers.

7.2 Analysis 2: Genre-Based Audio Profiling

7.2.1 Introduction

This analysis aims to understand the technical "signature" of different musical styles. It identifies how the structural components of music vary across the 114 identified genres.

7.2.2 General Description

A heatmap-style table was implemented to provide a high-density comparison of multiple variables. This allows for a quick visual scan to identify which genres are "outliers" in specific categories like Loudness or Acousticness.

7.2.3 Specific Requirements, Functions, and Formulas

- **Fields Configuration:** Track_Genre as rows, with aggregated measures as columns.
- **Measures Used:**
    - Avg.Accoust = AVERAGE('Tracks'[acousticness])
    - Avg.Instru = AVERAGE('Tracks'[instrumentalness])
    - Avg.Liveness = AVERAGE('Tracks'[liveness])
    - Avg.Loudness = AVERAGE('Tracks'[loudness])
    - Avg.Valence = AVERAGE('Tracks'[valence])

7.2.4 Analysis Results

- **High Energy Clusters:** Genres like **Party**, **Dancehall**, and **Metalcore** show the highest loudness and lowest acousticness.
- **Ambient/Classical Profiles: Classical** and **Sleep** genres are identified by extreme acousticness (0.92 and 0.66 respectively) and significantly lower loudness (negative decibel ranges like -20 to -24).
- **Mainstream Balance: Pop** exhibits mid-range scores across all metrics, suggesting a "safe" audio profile designed for wide accessibility.

7.2.5 Visualization

- **Visual Used:** Table with Conditional Formatting (Heatmap).

7.3 Analysis 3: Artist Performance Analysis

7.3.1 Introduction

This analysis evaluates individual artist success by aggregating their popularity across the entire dataset to find the most consistent high-performers.

7.3.2 General Description

A horizontal bar chart was used to rank artists based on their Avg.Popularity. This identifies market leaders and provides a benchmark for "top-tier" artist performance.

7.3.3 Specific Requirements

- **Axis:** Artists
- **Values:** Avg.Popularity

- **Tooltips:** Total_Track_Count (Used to distinguish between one-hit wonders and consistent performers).

7.3.4 Analysis Results

- **Market Leaders:** Artists such as **A.R. Rahman**, **Aespa**, and **Alan Walker** maintain average popularity scores in the **70+ range**.
- **Consistency:** Top-performing artists often have a Track Genre count of over 100, indicating they dominate multiple sub-genres while maintaining high popularity.

7.3.5 Visualization

- **Visual Used:** Clustered Bar Chart.

7.4 Analysis 4: Song Density (Danceability vs. Energy)

7.4.1 Introduction

This analysis explores how the library is distributed across different "vibes" by crossing two of the most important audio features: Danceability and Energy.

7.4.2 Visualization & Results

- **Visual Used:** Treemap.
- **Configuration:** Danceability_Bin and Energy_Bin as categories; Count of Tracks as values.
- **Insights:** The largest segment of the library falls under the **Medium Danceability / Medium Energy** category. The **High Danceability** segments represent a large portion of the tracks with low overall popularity, suggesting that while common, danceability alone does not guarantee a high popularity score.

7.5 Analysis 5: Tempo Trends and Musical Speed

7.5.1 Introduction

Tempo is a foundational element of music. This analysis examines how different "Tempo Bins" (Slow, Medium, Fast, etc.) impact both the danceability and the commercial success of a track.

7.5.2 Visualization & Results

- **Visual Used:** Line and Clustered Column Chart (Dual Axis).
- **X-axis:** Tempo_Bin (Medium, Fast, Slow, Very Fast, Very Slow, Extreme).
- **Y-axis:** Avg.Danceability (Line) and Avg.Popularity (Line).
- **Insights:**
    - **The Sweet Spot:** The **Slow** tempo bin actually shows a peak in **Avg.Popularity (approx. 54.8)** compared to other bins.
    - **Danceability Peak: Medium** tempo tracks exhibit the highest danceability (0.70), which then sharply declines as the tempo reaches "Extreme" levels.
    - **Inverse Relationship:** As tempo moves to **Extreme** speeds, popularity increases while danceability hits its lowest point (approx. 0.54).

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8. List of Analysis with Results

This section provides a **concise summary of all analyses performed** in the project along with their **key outcomes**. It serves as a quick reference for evaluators to understand what was analyzed and the major insights derived from each analytical objective.

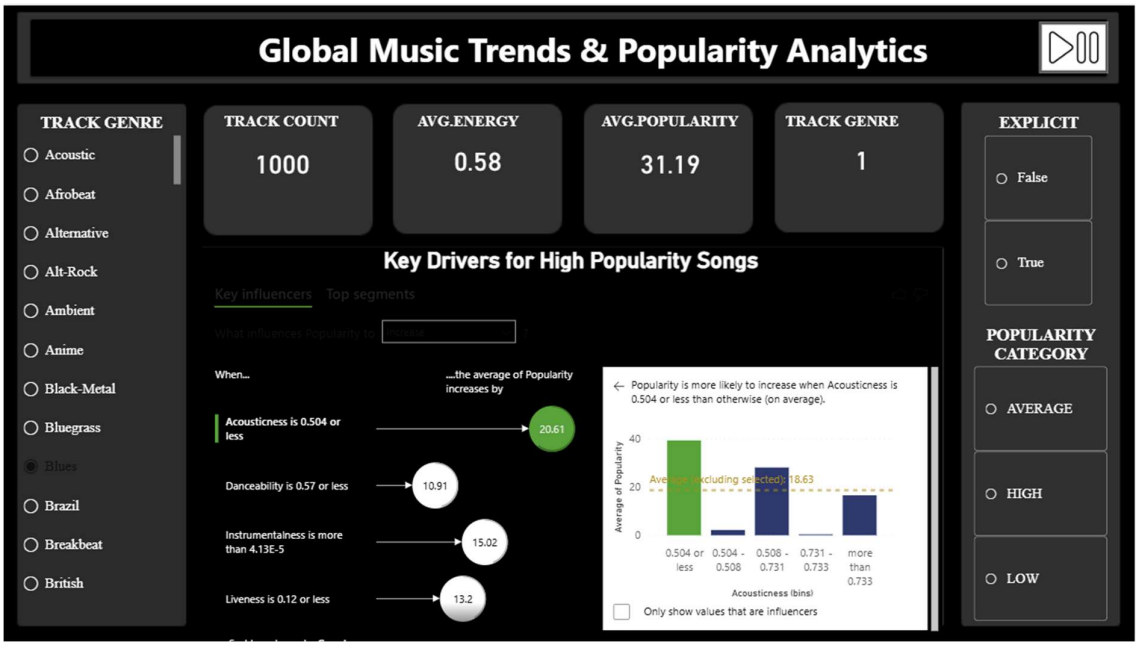
8.1 Summary of Analyses Performed

Analysis 1: Popularity Drivers Analysis (AI-Powered)

- **Objective:**
  Identify the key audio features that statistically influence a track's popularity.
- **Method Used:**
  Power BI Key Influencers AI visual.
- **Key Results:**
    - Low acousticness (≤ 0.504) increases popularity by approximately **20.61 units**.
    - Extremely low instrumentalness (vocal-heavy tracks) increases popularity by about **15.02 units**.

- Low liveness and specific danceability thresholds also contribute to popularity growth.

- **Conclusion:**
  Commercial success is strongly linked to **vocal dominance, controlled acoustic properties, and studio-quality production**.



Analysis 2: Genre-Based Audio Profiling

- **Objective:**
  Understand how audio characteristics differ across musical genres.

- **Method Used:**
  Heatmap-style table with conditional formatting.

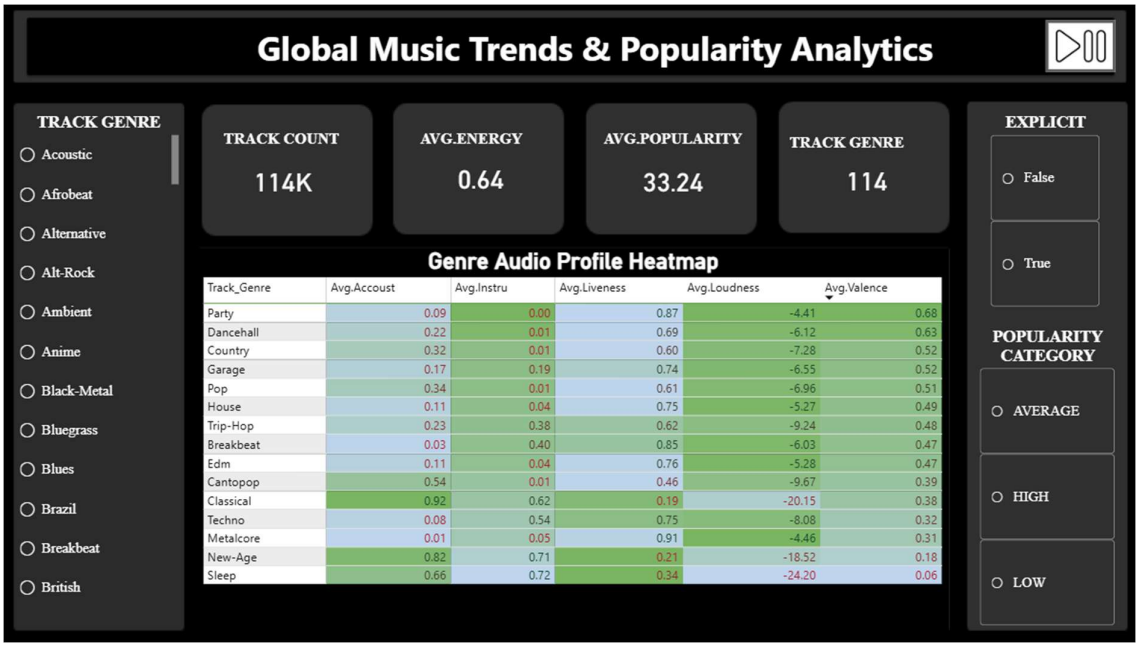
- **Key Results:**

  - High-energy genres (Party, Dancehall, Metalcore) show high loudness and low acousticness.

  - Classical and ambient genres exhibit very high acousticness and low loudness.

  - Pop music maintains balanced values across all features.

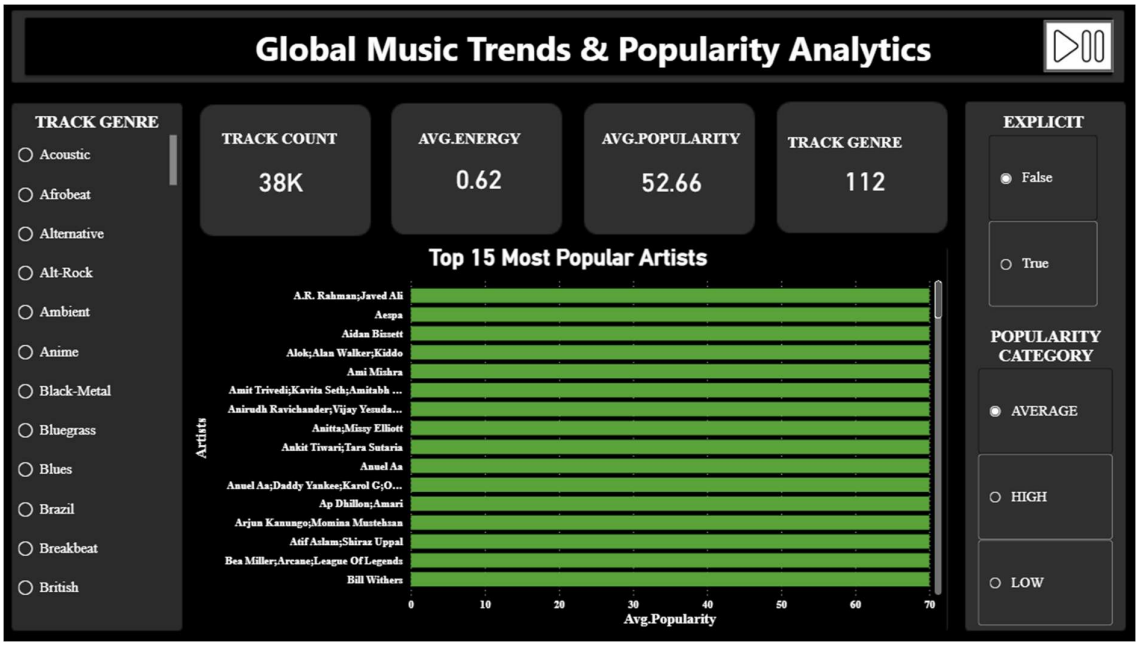
- **Conclusion:**
  Each genre has a distinct technical audio signature, and mainstream genres favor balanced audio profiles.



Track_Genre	Avg.Accoust	Avg.Instru	Avg.Liveness	Avg.Loudness	Avg.Valence
Party	0.09	0.00	0.87	-4.41	0.68
Dancehall	0.22	0.01	0.69	-6.12	0.63
Country	0.32	0.01	0.60	-7.28	0.52
Garage	0.17	0.19	0.74	-6.55	0.52
Pop	0.34	0.01	0.61	-6.96	0.51
House	0.11	0.04	0.75	-5.27	0.49
Trip-Hop	0.23	0.38	0.62	-9.24	0.48
Breakbeat	0.03	0.40	0.85	-6.03	0.47
Edm	0.11	0.04	0.76	-5.28	0.47
Cantopop	0.54	0.01	0.46	-9.67	0.39
Classical	0.92	0.62	0.19	-20.15	0.38
Techno	0.08	0.54	0.75	-8.08	0.32
Metalcore	0.01	0.05	0.91	-4.46	0.31
New-Age	0.82	0.71	0.21	-18.52	0.18
Sleep	0.66	0.72	0.34	-24.20	0.06

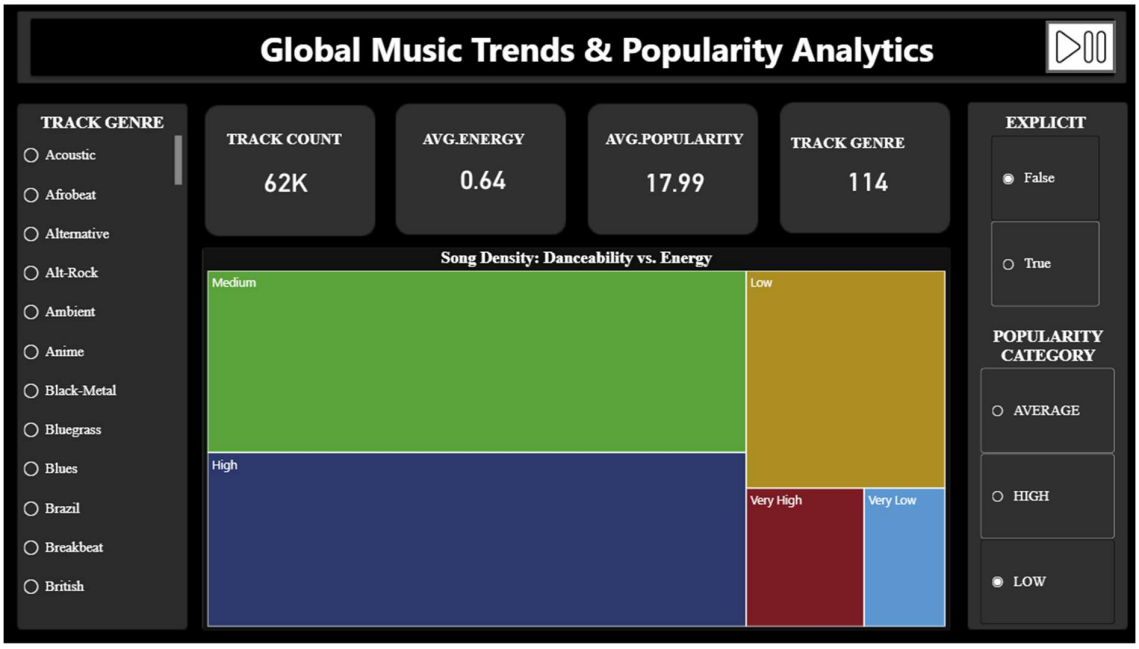
Analysis 3: Artist Performance Analysis

- **Objective:**
  Evaluate consistency and success of artists across the dataset.
- **Method Used:**
  Clustered bar chart showing average popularity by artist.
- **Key Results:**
  - Artists like **A.R. Rahman, Aespa, and Alan Walker** maintain average popularity above 70.
  - High-performing artists often have a large number of tracks, indicating sustained success.
- **Conclusion:**
  Long-term popularity is driven by **consistency and catalog strength**, not just isolated hits



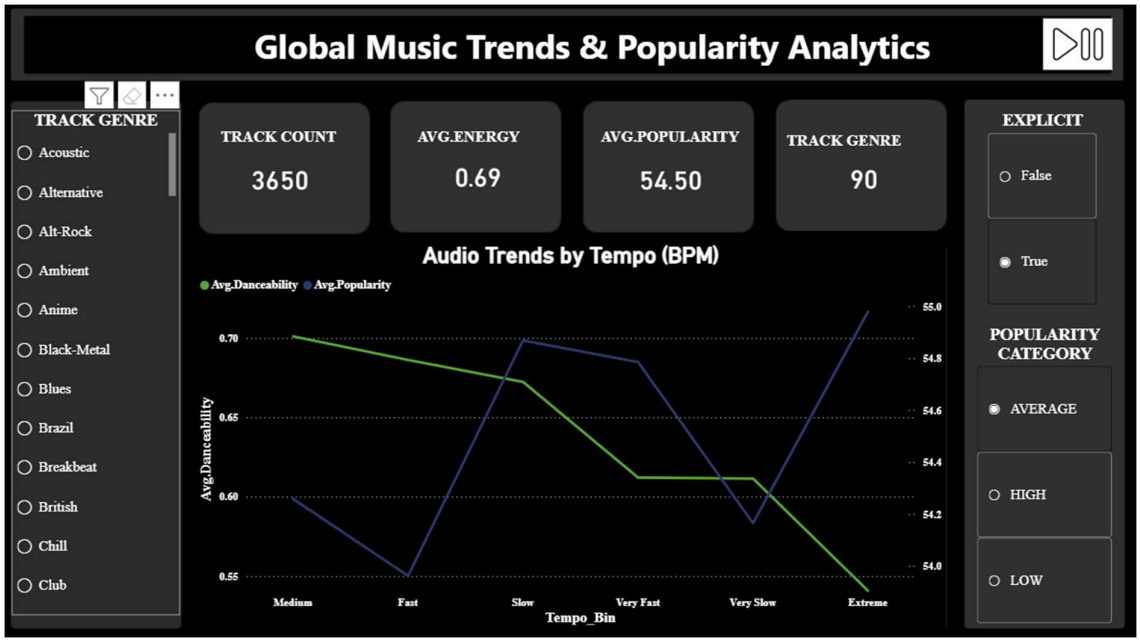
Analysis 4: Song Density Analysis (Danceability vs. Energy)

- **Objective:**
  Identify how songs are distributed across different musical "vibes".
- **Method Used:**
  Treemap using Danceability_Bin and Energy_Bin.
- **Key Results:**
  - Most tracks fall under **Medium Danceability / Medium Energy**.
  - High danceability alone does not guarantee high popularity.
- **Conclusion:**
  The music market is heavily saturated with mid-energy tracks, making differentiation essential for success.



Analysis 5: Tempo Trends and Musical Speed

- **Objective:**
  Examine the effect of tempo on popularity and danceability.
- **Method Used:**
  Dual-axis line and clustered column chart.
- **Key Results:**
  - Slow tempo tracks achieve the highest average popularity (~54.8).
  - Medium tempo tracks have the highest danceability (~0.70).
  - Extreme tempos show lower danceability but slightly higher popularity.
- **Conclusion:**
  Tempo has a **non-linear relationship** with popularity, revealing unexpected listener preferences.



8.2 Overall Analytical Findings

- Music popularity is influenced by **multiple interacting audio features**, not a single attribute.
- Balanced audio profiles tend to appeal to wider audiences.
- AI-driven analytics significantly enhance insight discovery in large datasets.
- Data-driven dashboards can effectively replace intuition-based decision-making in music analytics.

9.Future Scope:

The current project focuses on descriptive and diagnostic analytics to understand music popularity patterns using audio features. In the future, this work can be extended into **predictive analytics** by integrating machine learning models such as Linear Regression, Random Forest, or Gradient Boosting. By training these models on the existing dataset, the system could predict a song's potential popularity score before release, enabling artists, producers, and music labels to make data-driven decisions during the music production phase.

Another important extension of this project lies in **personalization and user-centric analysis**. By incorporating listener behavior data such as streaming history, playlist additions, and skip rates, the dashboard could evolve into a recommendation engine. Mapping user preferences to

audio feature profiles would allow the system to suggest songs and genres tailored to individual listeners, aligning the project more closely with real-world streaming platform applications.

Finally, the project can be scaled into an **enterprise-level analytics solution** by integrating external data sources and advanced deployment features. Adding social media engagement metrics, marketing data, and time-series information would enable trend forecasting and artist growth analysis. Publishing the dashboard to Power BI Service with automated refresh and security controls would make the solution suitable for use by music industry stakeholders such as record labels, streaming services, and talent management firms.

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